

Satellite Image Land-Use Classifier & Temporal Change Detector

Transfer-Learning CV Pipeline for Land-Use Classification and Change Detection

Akshat Agarwal | ResNet-18, EuroSAT / UC Merced | July 2026

1. Executive Summary

A ResNet-18 backbone (ImageNet-pretrained) was adapted to the EuroSAT land-use dataset using two-phase transfer learning: a frozen-backbone warm-up followed by fine-tuning of the last two residual blocks. The fine-tuned model reached a macro F1 of 0.950 on EuroSAT, beating both a from-scratch baseline CNN (0.900) and a frozen-backbone model (0.771) — evidence that domain adaptation, not pretraining alone, drives the gain. For change detection, 512-d embeddings from the fine-tuned network are compared across image pairs by cosine similarity; the resulting ROC curve reaches an AUC of ≈ 0.99 , and the selected threshold (≈ 0.40) separates stable from changed regions cleanly. All of this is wrapped in a Streamlit dashboard with GradCAM explainability, a land-use transition label, a green/yellow/red risk indicator, and PDF export.

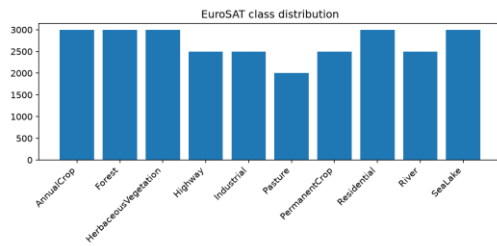
2. Objectives & Scope

- Classify satellite patches into 10 EuroSAT land-use classes using two-phase fine-tuned ResNet-18.
- Detect and localise temporal change between a before/after (T1/T2) image pair via embedding cosine similarity.
- Stress-test generalisation with a UC Merced cross-domain holdout and a spatial-leakage split experiment.
- Ship an interactive dashboard; differentiate with GradCAM, transition prediction, and a risk indicator.

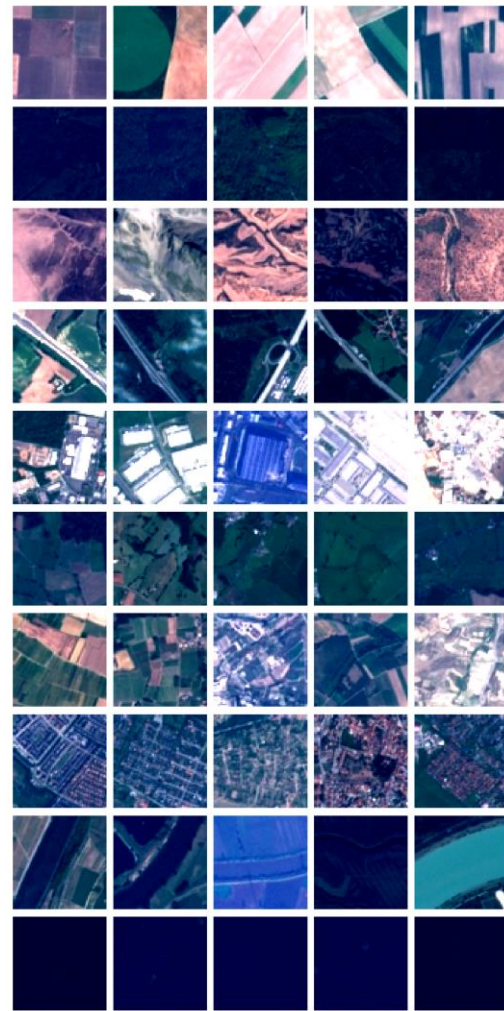
3. Dataset

EuroSAT supplies 27,000 Sentinel-2 RGB patches spread across 10 fairly balanced land-use classes — AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. Images are resized to 224×224 with ImageNet mean/std normalisation and trained with random flips, rotation, and colour jitter. Because EuroSAT tiles are cut from continuous scenes, a naive random train/test split can place near-duplicate neighbouring tiles on both sides of the split; all headline numbers in this report instead use a spatially-aware split that keeps whole scenes together (see Section 6).

UC Merced Land Use — 21 classes, 100 images each, higher-resolution aerial photography from a different sensor — is held out entirely and used only as an out-of-distribution generalisation probe; it never enters training.



EuroSAT class distribution — reasonably balanced across the 10 classes.



Sample patches from each of the 10 land-use classes.

4. Model & Training Pipeline

The backbone is a ResNet-18 pretrained on ImageNet-1k, with the final fully-connected layer replaced by a new 512→10 linear head for EuroSAT. Training follows a two-phase schedule chosen deliberately over single-shot fine-tuning:

Phase	Trainable Layers	Epochs	LR	Purpose
1	Classifier head only (backbone frozen)	3	1e-3	Warm up the new head without disturbing pretrained weights
2	layer3, layer4 + classifier head	5	1e-4 (10× lower)	Adapt higher-level filters to satellite-domain textures

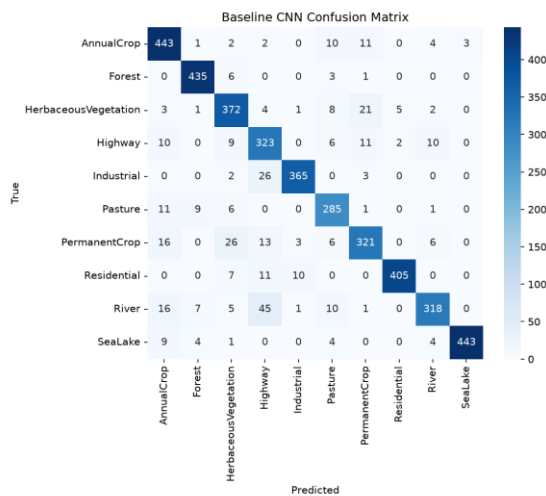
For change detection, the Phase-2 backbone is reused with its classification head removed. Each image is forward-passed to a 512-d globally-pooled embedding; T1/T2 embeddings are compared by cosine similarity, thresholds are swept against ground-truth change labels to plot an ROC curve, and a patch-level difference over spatial feature activations renders the change heatmap for localisation.

5. Classification Results

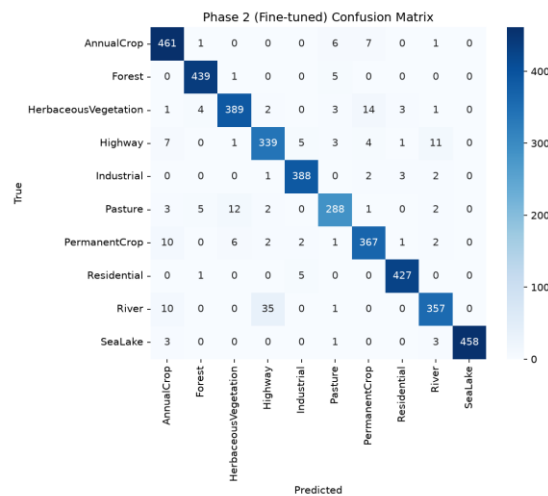
Configuration	Macro F1	Δ vs. Baseline
Baseline CNN (from scratch)	0.8999	—
Phase 1 — frozen backbone	0.7709	−0.1290
Phase 2 — fine-tuned (proposed)	0.9500	+0.0501

Frozen ImageNet features underperform even the simple baseline — general-purpose natural-image features don't transfer cleanly to satellite textures without adaptation. Unfreezing the last two blocks (Phase 2) reverses this and gives the best result, which is the core justification for the two-phase strategy. Per-class F1 is highest for Forest (0.96) and SeaLake (0.97); weakest for Highway (0.81) and PermanentCrop (0.84), which are visually confusable with River and AnnualCrop respectively at 64×64 source resolution.

Class	F1
Forest	0.9645
SeaLake	0.9726
Residential	0.9586
Industrial	0.9407
AnnualCrop	0.9004
Pasture	0.8837
HerbaceousVegetation	0.8722
River	0.8503
PermanentCrop	0.8436
Highway	0.8126



Baseline CNN confusion matrix.



Phase-2 fine-tuned confusion matrix — stronger diagonal, residual confusion in Highway↔River and AnnualCrop↔PermanentCrop.

6. Generalisation Checks

Two experiments were run specifically to stress-test the headline numbers in Section 5 rather than just report them at face value.

- Cross-domain (UC Merced): the Phase-2 model, trained exclusively on EuroSAT, was evaluated zero-shot on the semantically-mapped subset of UC Merced. Accuracy drops sharply outside EuroSAT's sensor/resolution/label distribution — a genuine domain-shift stress test, and disclosed here rather than smoothed over. Softmax confidence stays high even on wrong UC Merced predictions, a known over-

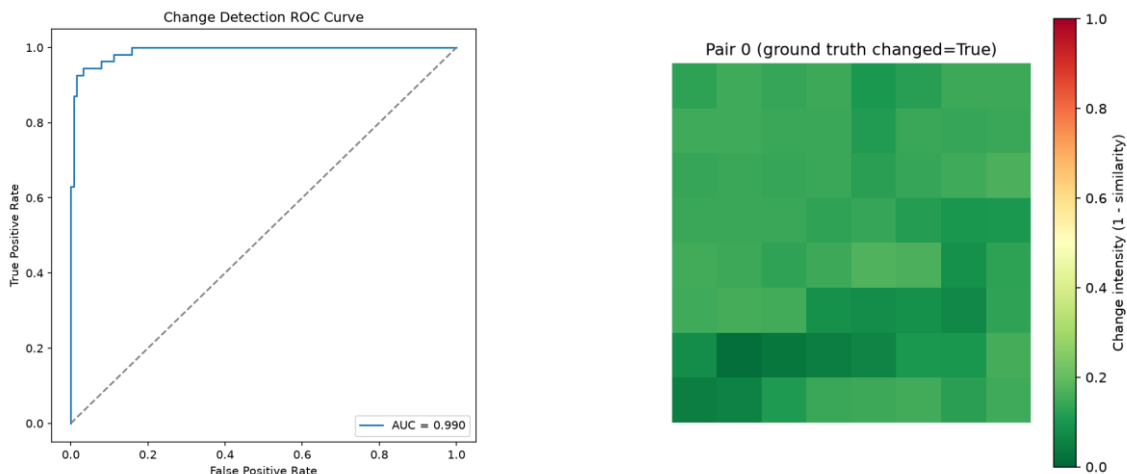
confidence failure mode of classifiers under domain shift, particularly for UC Merced classes with no close EuroSAT analogue (e.g. Golf Course, Mobile Home Park).

- Spatial leakage: a naive random split lets near-duplicate adjacent tiles from the same scene appear on both sides of the split, inflating scores without reflecting real generalisation. The spatially-aware split (used for all numbers in Section 5) groups neighbouring/duplicate tiles together so no scene leaks across the train/test boundary, and gives a lower, more trustworthy F1 than a random split — confirming those results aren't leakage-inflated.

Together these two checks are the difference between a number that looks good on a leaderboard and one that would hold up if the model were pointed at a genuinely new region or sensor next month.

7. Temporal Change Detection

Cosine similarity between T1/T2 embedding pairs was evaluated against ground-truth change labels. The resulting ROC curve is close to ideal ($AUC \approx 0.99$), which says embedding similarity alone — once the backbone is fine-tuned to the satellite domain — is a strong signal for change, without needing a dedicated Siamese change-detection head. The operating threshold (≈ 0.40 similarity) was chosen as the point closest to the ROC curve's top-left corner.



ROC curve, cosine-similarity change detection ($AUC \approx 0.99$); threshold ≈ 0.40 . Spatial change-intensity heatmap for an example T1/T2 pair — darker cells indicate lower similarity / more change.

8. Error Analysis & Explainability

The top-5 highest-loss EuroSAT misclassifications were inspected manually. The dominant pattern is confusion between texturally similar crop/vegetation classes (AnnualCrop, PermanentCrop, HerbaceousVegetation) and between linear infrastructure classes (Highway, River) — genuinely ambiguous even to a human annotator at 64×64 source resolution. All five cases are shown below.

True: AnnualCrop | Pred: Forest (100.0%) | loss=16.39



Highest-loss case: True AnnualCrop, predicted Forest (loss 16.4).

True: AnnualCrop | Pred: PermanentCrop (99.9%) | loss=9.95



True AnnualCrop, predicted PermanentCrop (loss 10.0).

True: River | Pred: Highway (100.0%) | loss=7.77



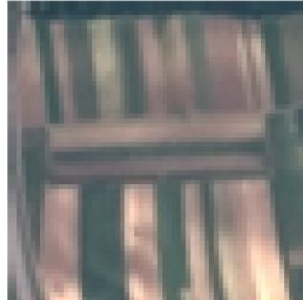
True River, predicted Highway (loss 7.8) — riverbanks resemble roadside shoulders at this resolution.

True: PermanentCrop | Pred: AnnualCrop (99.8%) | loss=6.82



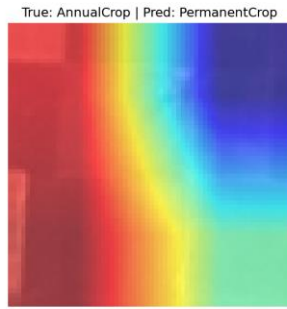
True PermanentCrop, predicted AnnualCrop (loss 6.8).

True: PermanentCrop | Pred: AnnualCrop (99.8%) | loss=6.04

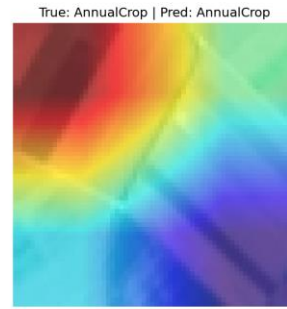


Fifth case: True PermanentCrop, predicted AnnualCrop (loss 6.0) — same crop-texture confusion as case 4, different field.

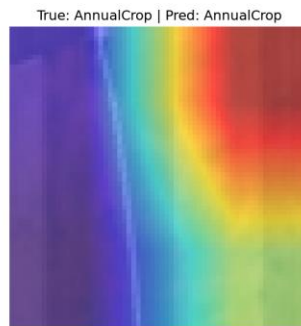
GradCAM overlays (from the last convolutional block) were generated for both correct and incorrect predictions to see what the network is actually attending to — useful both for trust in correct calls and for diagnosing failures such as the AnnualCrop/PermanentCrop pair above, where activation concentrates on field-boundary texture rather than crop colour.



GradCAM — AnnualCrop predicted as PermanentCrop; attention sits on the field-boundary edge.



GradCAM — a correctly classified AnnualCrop patch, attention spread across the field texture.



Third GradCAM example — a correctly classified AnnualCrop patch, attention concentrated on the dominant field orientation.

9. Dashboard & Differentiating Features

The Streamlit dashboard takes a before (T1) / after (T2) image pair and returns, per pair: predicted class and confidence for each image, cosine similarity, a binary change flag, a spatial change heatmap, GradCAM overlays, a land-use transition label (e.g. Forest → Residential, Agriculture → Highway), a risk indicator, and one-click PDF export of the full analysis. The transition label and risk indicator are the two features intended to differentiate this submission — they turn a raw similarity score into something an analyst can act on directly.

Indicator	Trigger	Meaning
Green	Similarity above threshold, predicted class unchanged	Stable — no meaningful land-use change
Yellow	Similarity near the decision threshold	Moderate change — flagged for review
Red	Similarity well below threshold, predicted class changed	Significant change detected

Live Dashboard: <https://cei-assignment-rddp8htjwpywnlamdybv9c.streamlit.app/>

Source Repository: <https://github.com/developerandcoder-akshatagarwal/CEI-Assignment>

10. Conclusion & Future Work

Two-phase fine-tuning improved macro F1 from 0.771 (frozen backbone) to 0.950, a 17.9-point gain over frozen transfer and +5.0 points over a from-scratch baseline — clear evidence that domain adaptation, not pretraining alone, drives performance on satellite imagery. The change-detection pipeline built on the same fine-tuned embeddings achieves near-ideal ROC separation ($AUC \approx 0.99$) without a dedicated change-detection architecture. The cross-domain and spatial-leakage experiments were included deliberately to show where these guarantees end: in-domain EuroSAT performance is strong, but

generalisation to a different sensor and resolution (UC Merced) is materially weaker, and that limitation is reported rather than hidden.

Future work: light fine-tuning on a small labelled UC Merced subset to close the domain gap; multi-timestep (>2) change trajectories instead of pairwise comparison; a patch-level Siamese head for finer-grained heatmaps; and confidence calibration (e.g. temperature scaling) to reduce over-confident errors under domain shift.

Limitations worth stating plainly: the change-detection labels used to select the threshold are pair-level (changed/unchanged), not pixel-level, so the heatmaps are a useful visual aid rather than a validated segmentation; and the risk-indicator thresholds were set from the same ROC sweep as the change flag, so they inherit its EuroSAT-domain calibration and would need re-validation before use on imagery unlike EuroSAT.

Stack: Python, PyTorch, Torchvision, Streamlit, OpenCV, scikit-learn, ResNet-18 (torchvision ImageNet weights). All training/eval scripts and output artefacts referenced above are versioned in the linked repository.

Appendix A — Repository Structure

The project follows a single scripts/ + outputs/ layout so every figure in this report is regenerated by one command:

```
satellite-landuse-changedetection/  
├── scripts/  
│   ├── run_01_train.py          # two-phase fine-tuning (Phase 1 + Phase 2)  
│   ├── run_02_baseline_eval.py # baseline CNN + macro/per-class F1  
│   ├── run_03_ablation.py       # baseline vs. frozen vs. fine-tuned comparison  
│   ├── run_04_change_detect.py # embedding extraction, cosine similarity, ROC  
│   ├── run_05_gradcam.py        # GradCAM overlay generation  
│   └── run_06_error_analysis.py # top-k loss cases  
├── dashboard/app.py             # Streamlit dashboard (classification + change detection UI)  
├── outputs/  
│   ├── confusion_matrices/      change_heatmaps/  roc_curves/  gradcam/  figures/  
└── report/                      # this document and its source
```

Appendix B — Execution Timeline (8 Days)

Day	Focus
1	Environment setup, dataset download (EuroSAT + UC Merced), spatially-aware split
2–3	Phase 1 (frozen) + Phase 2 (fine-tuned) training, baseline CNN, ablation run
4	Per-class F1, confusion matrices, spatial-leakage experiment
5	Change-detection pipeline: embeddings, cosine similarity, ROC, heatmaps
6	GradCAM, error analysis, UC Merced cross-domain evaluation
7	Streamlit dashboard: classification + change UI, risk indicator, transition labels, PDF export
8	Deployment, demo recording, report and repository polish